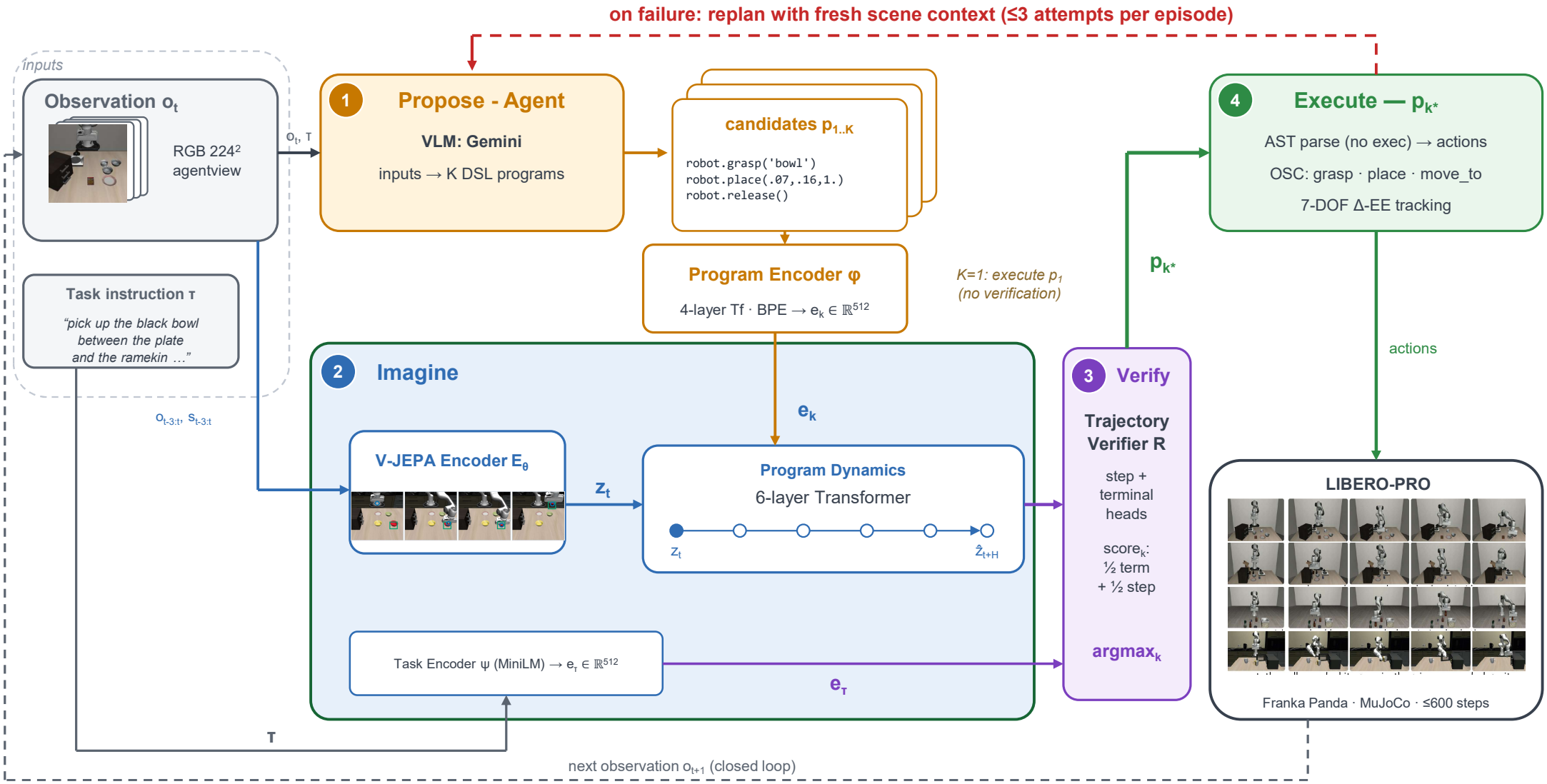

on failure: replan with fresh scene context (≤3 attempts per episode)
inputs
Observation o_t
RGB 224² agentview
o_t, τ
1 Propose - Agent
VLM: Gemini
inputs → K DSL programs
candidates $p_{1..K}$
robot.grasp('bowl')
robot.place(.07,.16,1.)
robot.release()
Program Encoder φ
4-layer Tf · BPE → $e_k \in \mathbb{R}^{512}$
K=1: execute p_1 (no verification)
4 Execute — p_{k^*}
AST parse (no exec) → actions
OSC: grasp · place · move_to
7-DOF Δ-EE tracking
p_{k^*}
actions
Task instruction τ
"pick up the black bowl between the plate and the ramekin …"
$o_{t-3:t}$, $s_{t-3:t}$
2 Imagine
e_k
V-JEPA Encoder E_θ
z_t
Program Dynamics
6-layer Transformer
z_t
$\hat{z}_{t+H}$
3 Verify
Trajectory Verifier R
step + terminal heads
$score_k$: ½ term + ½ step
$argmax_k$
LIBERO-PRO
Franka Panda · MuJoCo · ≤600 steps
Task Encoder ψ (MiniLM) → $e_\tau \in \mathbb{R}^{512}$
e_τ
τ
next observation o_{t+1} (closed loop)